

EduSaaS Additional Production Package

ER diagrams, DFD diagrams, seed data scripts, validation SQL, and deployment runbook in a single DOCX.

Scope: Complements the full database object catalog and PDDO documents.

1. Executive Summary

This document closes the remaining production-readiness gaps identified after the database object catalog review: graphical ER diagrams, graphical DFDs, seed scripts, validation SQL, and a deployment runbook. It is intended for architecture review, DBA handover, and deployment preparation.

Deliverable	Status
ER Diagram Images	Included
DFD Images	Included
Seed Data Scripts	Included
Validation SQL	Included
Deployment Runbook	Included

2. ER Diagram Images

Er Security

Security and Identity ER Diagram

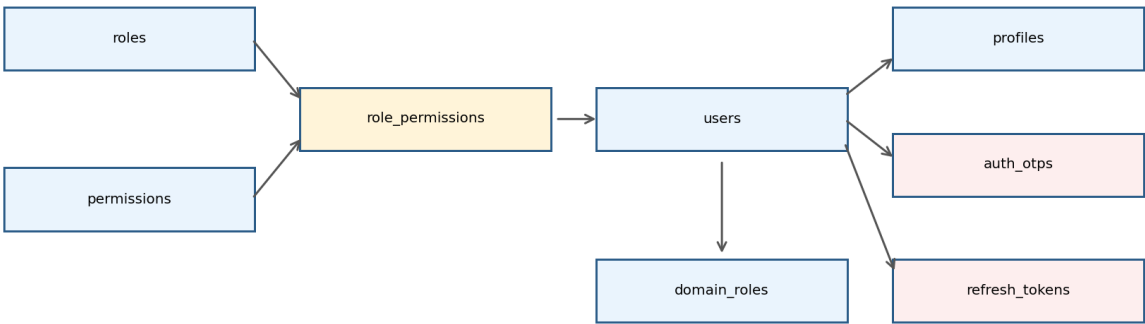


Figure: Er Security

Er Learning

Learning Domain ER Diagram

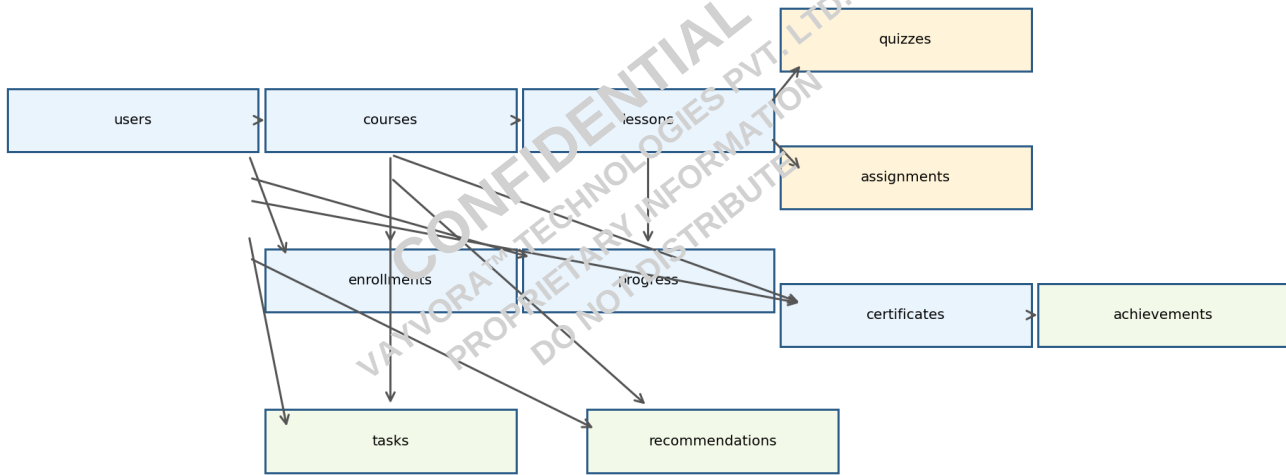


Figure: Er Learning

Er Assessment

Assessment and Skills ER Diagram

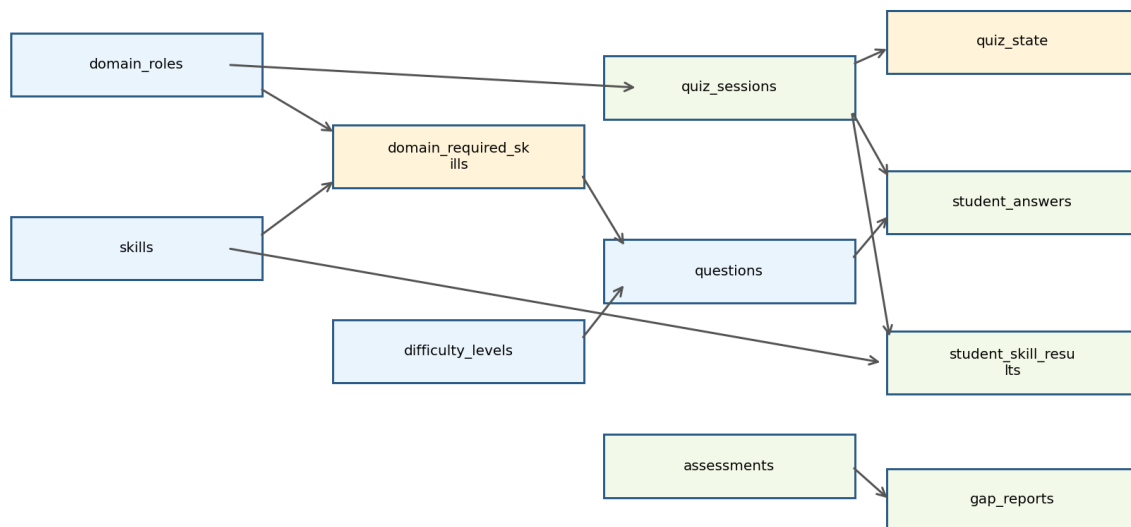


Figure: Er Assessment

Er Employer Community

Employer and Community ER Diagram

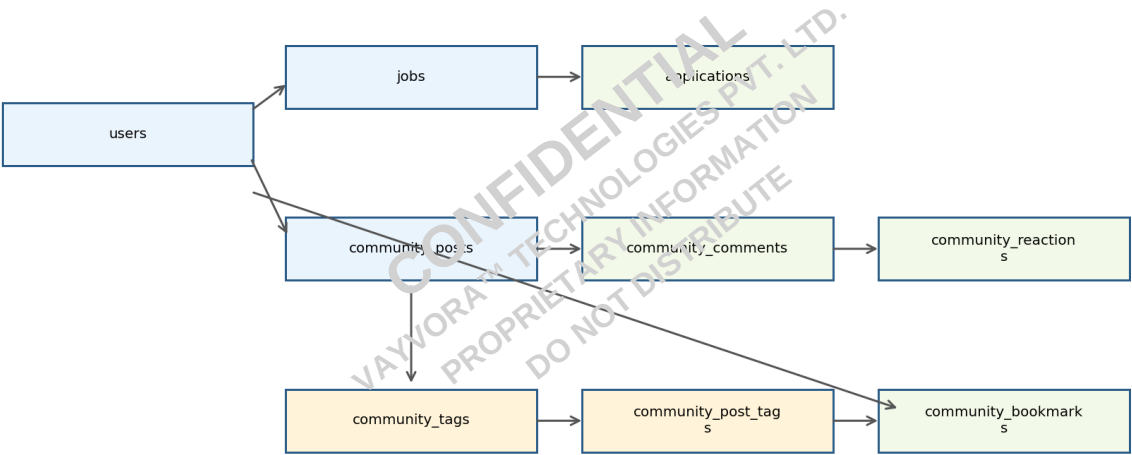


Figure: Er Employer Community

Er Admin Ai

Administration and AI Support ER Diagram

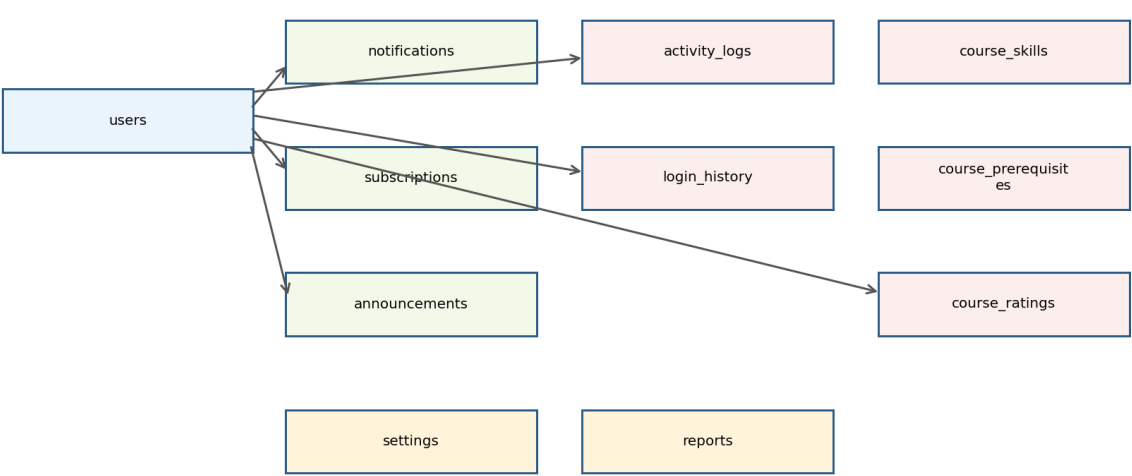


Figure: Er Admin Ai

3. Data Flow Diagram Images

Dfd Auth

DFD - Authentication and RBAC Flow



Figure: Dfd Auth

Dfd Learning

DFD - Learning Flow

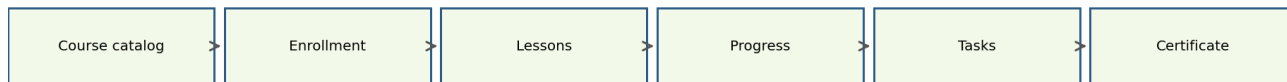


Figure: Dfd Learning

Dfd Assessment

DFD - Assessment Flow



Figure: Dfd Assessment

Dfd Ai

DFD - AI Recommendation Flow

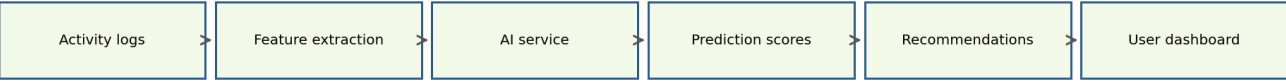


Figure: Dfd Ai

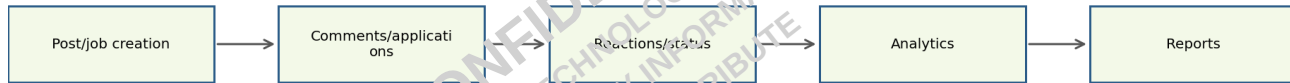


Figure: Dfd Community Employer

4. Seed Data Scripts

The following scripts seed minimum reference data required for RBAC, difficulty levels, domain roles, and platform settings. Review and adjust values before production execution.

4.1 seed_roles.sql

```

-- Seed roles
INSERT INTO education.roles (id, name, description)
VALUES
  (gen_random_uuid(), 'STUDENT', 'Learner user role'),
  (gen_random_uuid(), 'EDUCATOR', 'Course creator and instructor role'),
  (gen_random_uuid(), 'EMPLOYER', 'Employer and recruiter role'),
  (gen_random_uuid(), 'ADMIN', 'Platform administrator role'),
  (gen_random_uuid(), 'SUPER_ADMIN', 'Full platform owner role')
ON CONFLICT (name) DO NOTHING;

```

4.2 seed_permissions.sql

```

-- Seed permissions
INSERT INTO education.permissions (id, name, description, category)
VALUES
  (gen_random_uuid(), 'COURSE_VIEW', 'View course catalog and details', 'COURSE'),
  (gen_random_uuid(), 'COURSE_CREATE', 'Create courses', 'COURSE'),
  (gen_random_uuid(), 'COURSE_UPDATE', 'Update courses', 'COURSE'),
  (gen_random_uuid(), 'COURSE_DELETE', 'Delete courses', 'COURSE'),
  (gen_random_uuid(), 'LESSON_MANAGE', 'Create and maintain lessons', 'COURSE'),
  (gen_random_uuid(), 'ASSESSMENT_ATTEMPT', 'Attempt assessments and quizzes', 'ASSESSMENT'),
  (gen_random_uuid(), 'ASSESSMENT_REVIEW', 'Review assessment results', 'ASSESSMENT'),
  (gen_random_uuid(), 'JOB_CREATE', 'Create job postings', 'EMPLOYER'),
  (gen_random_uuid(), 'JOB_APPLY', 'Apply for jobs', 'EMPLOYER'),
  (gen_random_uuid(), 'APPLICATION_REVIEW', 'Review student applications', 'EMPLOYER'),
  (gen_random_uuid(), 'COMMUNITY_POST_CREATE', 'Create community posts', 'COMMUNITY'),
  (gen_random_uuid(), 'COMMUNITY_MODERATE', 'Moderate community content', 'COMMUNITY'),
  (gen_random_uuid(), 'USER_VIEW', 'View users', 'ADMIN'),
  (gen_random_uuid(), 'USER_UPDATE', 'Update users', 'ADMIN'),
  (gen_random_uuid(), 'REPORT_VIEW', 'View reports', 'REPORT'),
  (gen_random_uuid(), 'REPORT_EXPORT', 'Export reports', 'REPORT'),
  (gen_random_uuid(), 'SETTINGS_UPDATE', 'Update platform settings', 'ADMIN')
ON CONFLICT (name) DO NOTHING;

```

4.3 seed_role_permissions.sql

```

-- Seed role-permission mappings
INSERT INTO education.role_permissions (role_id, permission_id)
SELECT r.id, p.id
FROM education.roles r
JOIN education.permissions p ON p.name IN ('COURSE_VIEW', 'ASSESSMENT_ATTEMPT', 'JOB_APPLY', 'COMMUNITY_POST_CREATE')
WHERE r.name = 'STUDENT'
ON CONFLICT DO NOTHING;

```

```

INSERT INTO education.role_permissions (role_id, permission_id)
SELECT r.id, p.id
FROM education.roles r
JOIN education.permissions p ON p.name IN
('COURSE_VIEW', 'COURSE_CREATE', 'COURSE_UPDATE', 'LESSON_MANAGE', 'ASSESSMENT_REVIEW', 'COMMUNITY_POST_CREATE')
WHERE r.name = 'EDUCATOR'
ON CONFLICT DO NOTHING;

INSERT INTO education.role_permissions (role_id, permission_id)
SELECT r.id, p.id
FROM education.roles r
JOIN education.permissions p ON p.name IN ('JOB_CREATE', 'APPLICATION_REVIEW', 'COMMUNITY_POST_CREATE')
WHERE r.name = 'EMPLOYER'
ON CONFLICT DO NOTHING;

INSERT INTO education.role_permissions (role_id, permission_id)
SELECT r.id, p.id
FROM education.roles r
JOIN education.permissions p ON p.name IN ('USER_VIEW', 'USER_UPDATE', 'REPORT_VIEW', 'REPORT_EXPORT', 'COMMUNITY_MODERATE')
WHERE r.name = 'ADMIN'
ON CONFLICT DO NOTHING;

INSERT INTO education.role_permissions (role_id, permission_id)
SELECT r.id, p.id
FROM education.roles r
CROSS JOIN education.permissions p
WHERE r.name = 'SUPER_ADMIN'
ON CONFLICT DO NOTHING;

```

4.4 seed_reference_data.sql

```

-- Seed difficulty levels
INSERT INTO education.difficulty_levels (difficulty_name, difficulty_order)
VALUES
    ('Easy', 1),
    ('Medium', 2),
    ('Hard', 3),
    ('Expert', 4)
ON CONFLICT (difficulty_name) DO NOTHING;

-- Seed sample domain roles
INSERT INTO education.domain_roles (domain_role_id, domain_name, category)
VALUES
    (gen_random_uuid(), 'Java Developer', 'Software'),
    (gen_random_uuid(), 'Data Scientist', 'AI'),
    (gen_random_uuid(), 'Cloud Engineer', 'Cloud'),
    (gen_random_uuid(), 'DevOps Engineer', 'Cloud'),
    (gen_random_uuid(), 'Business Analyst', 'Business')
ON CONFLICT (domain_name) DO NOTHING;

-- Seed platform settings
INSERT INTO education.settings (id, scope, key, value)
VALUES
    (gen_random_uuid(), 'system', 'RECOMMENDATION_LIMIT', '{"value":10}::jsonb'),
    (gen_random_uuid(), 'assessment', 'DEFAULT_QUIZ_QUESTIONS', '{"value":50}::jsonb'),
    (gen_random_uuid(), 'notification', 'NOTIFICATION_RETENTION_DAYS', '{"value":365}::jsonb')
ON CONFLICT (scope, key) DO NOTHING;

```

5. Validation SQL

Run these queries after deployment to validate the object catalog, constraints, indexes, seed data, and AI support objects.

```

-- 01_validate_table_count.sql
SELECT COUNT(*) AS table_count
FROM information_schema.tables
WHERE table_schema = 'education'
AND table_type = 'BASE TABLE';

-- 02_validate_enums.sql
SELECT n.nspname AS schema_name, t.typname AS enum_name
FROM pg_type t
JOIN pg_namespace n ON n.oid = t.typnamespace
WHERE n.nspname = 'education'
AND t.typtype = 'e'
ORDER BY t.typname;

```

```

-- 03_validate_indexes.sql
SELECT schemaname, tablename, indexname
FROM pg_indexes
WHERE schemaname = 'education'
ORDER BY tablename, indexname;

-- 04_validate_foreign_keys.sql
SELECT tc.table_schema, tc.table_name, tc.constraint_name
FROM information_schema.table_constraints tc
WHERE tc.table_schema = 'education'
  AND tc.constraint_type = 'FOREIGN KEY'
ORDER BY tc.table_name, tc.constraint_name;

-- 05_validate_primary_keys.sql
SELECT tc.table_schema, tc.table_name, tc.constraint_name
FROM information_schema.table_constraints tc
WHERE tc.table_schema = 'education'
  AND tc.constraint_type = 'PRIMARY KEY'
ORDER BY tc.table_name;

-- 06_validate_unique_constraints.sql
SELECT tc.table_schema, tc.table_name, tc.constraint_name
FROM information_schema.table_constraints tc
WHERE tc.table_schema = 'education'
  AND tc.constraint_type = 'UNIQUE'
ORDER BY tc.table_name, tc.constraint_name;

-- 07_validate_functions.sql
SELECT routine_schema, routine_name, routine_type
FROM information_schema.routines
WHERE routine_schema = 'education'
ORDER BY routine_name;

-- 08_validate_triggers.sql
SELECT trigger_schema, event_object_table, trigger_name, event_manipulation
FROM information_schema.triggers
WHERE trigger_schema = 'education'
ORDER BY event_object_table, trigger_name;

-- 09_validate_seed_roles.sql
SELECT name FROM education.roles ORDER BY name;

-- 10_validate_seed_permissions.sql
SELECT category, COUNT(*) AS permission_count
FROM education.permissions
GROUP BY category
ORDER BY category;

-- 11_validate_ai_tables.sql
SELECT table_name
FROM information_schema.tables
WHERE table_schema = 'education'
  AND table_name IN ('activity_logs', 'login_history', 'course_skills', 'course_prerequisites', 'course_ratings')
ORDER BY table_name;

```

6. Deployment Runbook

Step	Phase	Action
1	Pre-check	Confirm backup, connection, target database, roles, and maintenance window approval.
2	Schema and enums	Run schema and enum scripts before tables.
3	Baseline tables	Create all 41 baseline tables from Edu_saas_DB_Scripts.sql.
4	Baseline indexes	Create extracted baseline indexes.
5	Baseline FKs	Apply extracted baseline foreign key ALTER statements.
6	AI support tables	Create activity_logs, login_history, course_skills, course_prerequisites, and course_ratings.
7	Recommendation extensions	Alter recommendations table to add AI scoring/ranking columns.
8	AI indexes	Create AI and recommendation support indexes.
9	Functions and triggers	Create set_updated_at() and updated_at triggers for AI support tables and recommendations.
10	Seed data	Run seed roles, permissions, role_permissions,

		difficulty levels, domain roles, and settings.
11	Validation	Run validation SQL package and capture evidence.
12	Smoke test	Validate login role lookup, course query, enrollment insert in lower environment, recommendation query, and reporting query.
13	Sign-off	DBA, application owner, security, QA, and architecture review sign-off.

6.1 Rollback Runbook

```
-- Rollback order should be reverse of deployment.
-- 1. Drop AI triggers.
DROP TRIGGER IF EXISTS trg_recommendations_updated_at ON education_recommendations;
DROP TRIGGER IF EXISTS trg_course_ratings_updated_at ON education.course_ratings;
DROP TRIGGER IF EXISTS trg_course_prerequisites_updated_at ON education.course_prerequisites;
DROP TRIGGER IF EXISTS trg_course_skills_updated_at ON education.course_skills;
DROP TRIGGER IF EXISTS trg_login_history_updated_at ON education.login_history;
DROP TRIGGER IF EXISTS trg_activity_logs_updated_at ON education.activity_logs;

-- 2. Drop AI function if no longer required.
DROP FUNCTION IF EXISTS education.set_updated_at();

-- 3. Drop AI indexes.
DROP INDEX IF EXISTS education.idx_recommendations_user_rank;
DROP INDEX IF EXISTS education.idx_recommendations_confidence_score;
DROP INDEX IF EXISTS education.idx_recommendations_rank;
DROP INDEX IF EXISTS education.idx_course_ratings_rating;
DROP INDEX IF EXISTS education.idx_course_ratings_course_id;
DROP INDEX IF EXISTS education.idx_course_ratings_user_id;
DROP INDEX IF EXISTS education.idx_course_prerequisites_prerequisite_course_id;
DROP INDEX IF EXISTS education.idx_course_prerequisites_course_id;
DROP INDEX IF EXISTS education.idx_course_skills_skill_id;
DROP INDEX IF EXISTS education.idx_course_skills_course_id;
DROP INDEX IF EXISTS education.idx_login_history_days_since_last_login;
DROP INDEX IF EXISTS education.idx_login_history_user_id;
DROP INDEX IF EXISTS education.idx_activity_logs_user_last_activity;
DROP INDEX IF EXISTS education.idx_activity_logs_last_activity;
DROP INDEX IF EXISTS education.idx_activity_logs_user_id;

-- 4. Drop AI support tables only if rollback scope requires removing AI feature completely.
DROP TABLE IF EXISTS education.course_ratings;
DROP TABLE IF EXISTS education.course_prerequisites;
DROP TABLE IF EXISTS education.course_skills;
DROP TABLE IF EXISTS education.login_history;
DROP TABLE IF EXISTS education.activity_logs;
```

7. Final Production Checklist

Checklist Item	Status
Backup taken before deployment	Required
Schema and enums deployed	Required
46 tables available	Required
Foreign keys valid	Required
Indexes created	Required
Seed roles and permissions loaded	Required
AI tables created if AI feature is in scope	Required
Validation SQL executed	Required
Smoke tests passed	Required
Rollback tested in lower environment	Required
Architecture/DBA/Security approval captured	Required

8. Notes and Assumptions

The diagrams are logical diagrams intended for PDDD and architecture review. Physical database DDL remains in the Full Database Object Catalog document. Seed data values should be reviewed and adjusted by product/security owners before production execution.